US 20210064985A1

(19) **United States**

(12) **Patent Application Publication** (10) **Pub. No.: US 2021/0064985 A1**

Sun et al. (43) **Pub. Date: Mar. 4, 2021**

(54) **MACHINE LEARNING HARDWARE HAVING REDUCED PRECISION PARAMETER COMPONENTS FOR EFFICIENT PARAMETER UPDATE**

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(21) Appl. No.: **16/558,585**

(22) Filed: **Sep. 3, 2019**

Publication Classification

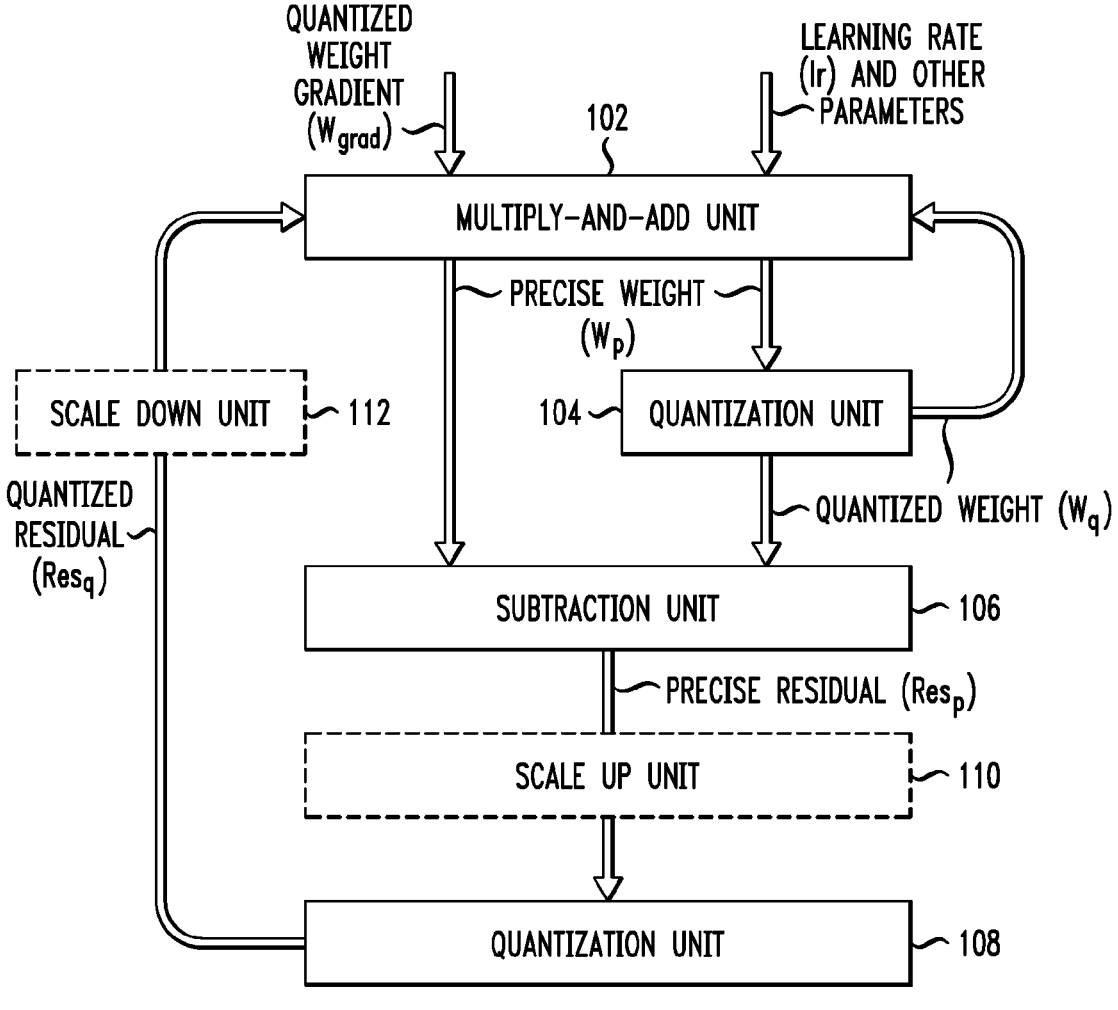
(51) **Int. Cl.**

G06N 3/08 (2006.01)

G06N 3/04 (2006.01)

(52) **U.S. Cl.**

CPC ***G06N 3/08*** (2013.01); ***G06N 3/04*** (2013.01)

(57) **ABSTRACT**

An apparatus for training and inferencing a neural network includes circuitry that is configured to generate a first weight having a first format including a first number of bits based at least in part on a second weight having a second format including a second number of bits and a residual having a third format including a third number of bits. The second number of bits and the third number of bits are each less than the first number of bits. The circuitry is further configured to update the second weight based at least in part on the first weight and to update the residual based at least in part on the updated second weight and the first weight. The circuitry is further configured to update the first weight based at least in part on the updated second weight and the updated residual.